Practical : 9

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Problem Statement: String Processing: Naïve String Matching - CO1, CO2, CO4

Given: l A text string text of length n. l A pattern string pattern of length m. Objective: Find all starting indices i in the text such that the substring text[i:i+m] is exactly equal to the

pattern pattern, using the Naive String Matching Algorithm approach. Constraints: l 0 ≤ m ≤ n l Characters in text and pattern can be any valid characters (e.g., a–z, A–Z, digits, etc.)

Code

```
# Naive String Matching Algorithm


text = input("Enter Text: ")

pattern = input("Enter Pattern: ")


n = len(text)

m = len(pattern)


found = False


print("\nPattern found at index:")


for i in range(n - m + 1):


    if text[i:i+m] == pattern:

        print(i)

        found = True
```

```
if not found:
    print("Pattern not found")
```

Output 1

Input

```
Enter Text: AABAACAADAABAABA

Enter Pattern: AABA
```

Output

```
Pattern found at index:

0

9

12
```

OR

Output 2

Input

```
Enter Text: COMPUTERENGINEERING

Enter Pattern: ENG
```

Output

```
Pattern found at index:

8
```

OR

Output 3 (Pattern Not Found)

Input

```
Enter Text: PYTHONPROGRAM
```

```
Enter Pattern: JAVA
```

Output

```
Pattern found at index:

Pattern not found
```